

Do Small Language Models Know When They Are Wrong?

Internal-State Probing for Hallucination Detection in Sub-4B Language Models, with a Study of Cross-Model and Cross-Lingual Transfer

Project type	Final-year major project (research-oriented)
Area	Machine Learning · Natural Language Processing · Model Interpretability
Duration	24 weeks
Compute budget	Google Colab (free tier) / Kaggle — single 15 GB GPU
Target venue	IEEE / Scopus-indexed conference (+ arXiv preprint)
Primary deliverables	Research paper, open-source codebase, interactive demo

Prepared as a complete end-to-end technical specification

Abstract

Large language models frequently produce fluent, confident, and factually incorrect output — a failure mode known as *hallucination*. Current detection strategies are predominantly *output-based*: they sample the model many times and measure disagreement, or they verify claims against an external corpus. Both are expensive, and both treat the model as a black box.

This project investigates an alternative: that the information required to detect a hallucination is *already present inside the model's own forward pass*, and can be extracted by a classifier several orders of magnitude cheaper than sampling-based methods. Prior work has established that such signals exist in 7B–70B parameter models. This project asks three questions that remain open, and answers them in the small-model regime (1B–4B parameters) that is relevant to on-device deployment:

(RQ1) Does a linearly decodable truthfulness signal exist in sub-4B models at all? **(RQ2)** Does a detector trained on one model family transfer to a *different* model family under a learned representational alignment? **(RQ3)** Is the signal encoded language-independently, such that an English-trained detector transfers to Indic-language prompts?

The evaluation departs from the AUROC-only convention of the probing literature and adopts a *selective-prediction* protocol, reporting risk–coverage curves and plotting detection quality against inference cost. The central practical claim to be tested is that an internal-state probe can approach the detection quality of semantic entropy at roughly one order of magnitude less compute. Every experiment is designed to run within a free-tier Colab allocation via a one-time activation-caching strategy.

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1 Introduction and Motivation

1.1 The hallucination problem

A language model trained by next-token prediction has no mechanism that distinguishes *recalling* a fact from *fabricating* a plausible continuation. Both are simply high-probability sequences. The consequence is that a model will answer a question about a non-existent citation, a fictional API method, or an obscure biographical detail with exactly the same fluency and syntactic confidence it applies to well-attested facts. The surface form of the output carries almost no diagnostic information about its reliability.

This is the primary blocker to deploying language models in any setting where a wrong answer is more costly than no answer — clinical decision support, legal drafting, financial summarisation, code generation against unfamiliar libraries. In every one of these settings, a system that says *"I do not know"* 15% of the time and is reliable the other 85% is strictly more valuable than a system that always answers and is wrong 15% of the time. Converting the second system into the first requires exactly one missing component: a detector.

1.2 Why existing detectors are unsatisfying

The dominant families of detection methods share a structural weakness — they operate on the model's *outputs*, and pay a multiplicative compute cost to do so.

Family	Mechanism	Cost and limitation
Sampling / consistency (SelfCheckGPT, semantic entropy)	Generate k independent answers; measure semantic disagreement between them	$k\times$ generation cost (typically $k = 5\text{--}20$), plus an NLI model for clustering. Latency becomes unacceptable for interactive use
Retrieval verification	Retrieve supporting evidence; check entailment of each claim	Requires a trustworthy corpus and a retrieval stack. Cannot verify what is not indexed
Verbalised confidence ($P(\text{True})$)	Ask the model to rate its own confidence	Cheap, but poorly calibrated; inherits the same generative process that produced the error
Token likelihood	Use sequence log-probability or entropy as a proxy	Nearly free, but conflates <i>linguistic</i> uncertainty (many valid phrasings) with <i>factual</i> uncertainty

Table 1. Output-based hallucination detection families and their characteristic costs.

Every one of these methods discards something the model has already computed. A forward pass through a 3B-parameter transformer produces on the order of tens of thousands of intermediate activation values before a single token is emitted. Output-based methods throw all of that away and then pay to reconstruct an uncertainty signal from the outside.

1.3 The central intuition

CORE HYPOTHESIS

The computation that distinguishes *retrieval from memory* from *plausible fabrication* leaves a trace in the model's intermediate representations. That trace is linearly decodable, and it is available **before or during** generation — at the cost of a single dot product.

The analogy that makes this concrete is a polygraph. A subject's spoken words are a *deliberate, optimised output*; their physiological signals are not. A transformer's emitted tokens are optimised for likelihood under the training distribution; its residual stream is merely the machinery that produced them, and may retain information the output-selection process discarded.

This intuition is not new, and this project does not claim it. It has been demonstrated in models of 7B parameters and above by several independent groups (Section 4). What has not been established is whether it survives at the parameter scales that matter for practical deployment, and — more interestingly — whether the learned detector is *portable* across models. Those are this project's questions.

2 Problem Statement and Scope

2.1 Formal problem definition

Let M be an autoregressive language model with L transformer layers and hidden dimension d . For an input prompt x , let $y = M(x)$ denote the generated answer, and let

$$\mathbf{h}_\ell^{(t)}(x, y) \in \mathbb{R}^d, \quad \ell \in \{1, \dots, L\}, \quad t \in \text{token positions}$$

denote the residual-stream activation at layer ℓ and token position t . Let $c(x, y) \in \{0, 1\}$ be a ground-truth correctness label, where 1 indicates the generated answer is factually incorrect (a hallucination).

We seek a probe g_θ operating on a selected (layer, position) pair — or a small set of them — such that

$$g_\theta(\mathbf{h}_{\ell^*}^{(t^*)}) \approx \Pr[c(x, y) = 1] \tag{1}$$

subject to three constraints that define the project's character:

1. **Cost.** Evaluating g_θ must add negligible cost over the single forward pass already required to produce y — ruling out any method that samples the model repeatedly.
2. **Scale.** M is restricted to models with fewer than 4B parameters, runnable in 4-bit quantisation on a single 15 GB consumer GPU.
3. **Generalisation.** g_θ is evaluated on held-out *entities and topics*, and additionally under distribution shift across models, tasks, and languages.

2.2 In scope and out of scope

In scope	Explicitly out of scope
Detection of factual errors in short-form generative QA	Correcting or repairing hallucinations (intervention)
Statement-level truthfulness classification	Fine-tuning, RLHF, or any weight modification of M
Models in the 1B–4B range, 4-bit quantised	Models above ~8B (compute-infeasible here)
Cross-model, cross-task, cross-lingual transfer of probes	Multimodal (vision–language) hallucination
Selective prediction / abstention as the deployment framing	Long-form, multi-claim factuality decomposition

Table 2. Scope boundaries. The out-of-scope column exists to keep a 24-week project finishable.

3 Technical Background

3.1 The transformer residual stream

A decoder-only transformer maintains, for each token position, a d -dimensional vector that is progressively refined by each layer. Layer ℓ reads this vector, computes an attention update and an MLP update, and *adds* both back:

$$\mathbf{h}_\ell = \mathbf{h}_{\ell-1} + \text{Attn}_\ell(\mathbf{h}_{\ell-1}) + \text{MLP}_\ell(\mathbf{h}_{\ell-1}) \quad (2)$$

Because updates are additive, $\mathbf{h}_\ell$ is best understood not as a transformed copy of the input but as a shared communication channel — the *residual stream* — onto which successive layers write features. This is the property that makes probing viable: features written by early and middle layers remain linearly accessible at later layers, so a linear read-out at a well-chosen depth can recover information the final output distribution does not surface.

Empirically, the literature reports a consistent depth profile: early layers encode surface lexical and syntactic properties; middle-to-late layers (roughly 55–80% of depth) encode semantic and factual properties; the final layers specialise toward next-token prediction and partially discard abstract features. We therefore expect the truthfulness signal to peak in the upper-middle band, and one of the first empirical results this project produces is a direct measurement of that profile for small models.

3.2 What a probe is

A *probe* is a small supervised classifier trained on frozen internal activations. The model under study is never updated. In its simplest and most important form the probe is logistic regression:

$$p(\text{hallucination}) = \sigma(\mathbf{w}^\top \mathbf{h} + b), \quad \sigma(z) = (1 + e^{-z})^{-1} \quad (3)$$

The parameter count is $d + 1$ — roughly 2,000 numbers. Training takes seconds on a CPU. The deliberate simplicity is methodological, not merely practical: if a *linear* read-out succeeds, the claim "this information is explicitly represented" is defensible. If only a deep non-linear probe succeeds, the probe may be computing the answer itself rather than reading it — the well-known probe-expressivity confound. We therefore treat the linear probe as the primary result and non-linear probes as a controlled comparison.

3.3 The linear representation hypothesis

Underlying this design is the hypothesis that high-level concepts are encoded as approximately linear directions in activation space, so that a concept can be read by projection onto a single vector. Evidence spans classical word-embedding analogies through to explicit demonstrations of linear truth structure in LLaMA-family representations. This project does not aim to settle the hypothesis; it adopts it as a working assumption and reports where it holds and where it fails at small scale.

3.4 A taxonomy of hallucination

Precision here matters, because different subtypes are detectable by different mechanisms and conflating them produces uninterpretable results.

Type	Definition	Treatment in this project
Factuality	Output contradicts established world knowledge	Primary target. Main evaluation setting
Faithfulness	Output contradicts a provided source document	Secondary; used for the cross-task transfer experiment
Knowledge-gap	Model genuinely never learned the fact	Expected to be <i>detectable</i> — internal state should reflect absence of retrieval
Retrieval-failure	Fact was learned but not surfaced under this phrasing	Expected to be <i>harder</i> ; analysed separately in failure analysis
Reasoning error	Premises correct, inference invalid	Out of scope; excluded from primary datasets

Table 3. Hallucination taxonomy and its role in the experimental design.

WHY THIS DISTINCTION EARNS CREDIT

A probe that separates knowledge-gap errors well but retrieval-failure errors poorly is a scientifically interesting result, not a failure. Reporting the breakdown — rather than a single pooled AUROC — is one of the concrete ways this work will be more informative than a straightforward replication.

4 Literature Review

4.1 Internal-state and probing methods

Azaria and Mitchell [1] introduced SAPLMA, training a feed-forward classifier on the hidden activation of the final token of a declarative statement to predict its truth value. They established the middle-to-late layer depth profile and released topic-partitioned true/false datasets that remain a standard warm-up benchmark.

Burns et al. [2] proposed Contrast-Consistent Search (CCS), recovering a truth direction *without labels* by exploiting the logical constraint that a statement and its negation must receive complementary probabilities. This matters as a control: if an unsupervised direction recovers much of the supervised probe's performance, the feature is intrinsic to the model rather than an artefact of label fitting.

Li et al. [3] (Inference-Time Intervention) localised truthfulness to a sparse set of attention heads and showed that shifting activations along the identified direction improves TruthfulQA performance — causal evidence that the direction is used by the model, not merely correlated with truth.

Marks and Tegmark [4] characterised the geometry of the truth direction in LLaMA representations, finding approximately linear structure that generalises across topic datasets, and provided causal evidence via activation patching.

Chen et al. [5] (INSIDE) moved from statements to generations, proposing EigenScore — the log-determinant of the covariance of internal states across multiple sampled responses — as a hallucination score. This is directly relevant but reintroduces a sampling cost, and is therefore treated here as a strong baseline rather than a competing framing.

CH-Wang et al. [6] probed for hallucination during open-ended generation rather than on curated statements, and reported that probe performance degrades substantially in the generative setting relative to the statement setting.

Orgad et al. [7] is the most important related work for this proposal, and the one that most constrains it. They report three findings: (i) the truthfulness signal is concentrated in the *exact answer tokens* rather than the final token of the sequence; (ii) probes generalise poorly across task types within a model, suggesting truthfulness is encoded in a skill-specific rather than universal manner; (iii) internal states can predict the specific error type. Finding (ii) directly weakens a naive cross-task framing.

CONSEQUENCE FOR THIS PROPOSAL — READ CAREFULLY

Because [7] has already shown limited *cross-task* generalisation within a model, this project must not present cross-task transfer as its headline novelty. It is retained as a *secondary, confirmatory* experiment at small scale. The headline novelty is relocated to **cross-model-family** transfer (RQ2) and **cross-lingual** transfer (RQ3), neither of which [7] or any other cited work addresses. Being explicit about this in the paper is what separates an informed contribution from an accidental replication.

4.2 Output-based uncertainty methods

Kadavath et al. [8] established that models can be prompted to estimate $P(\text{True})$ for their own answers with non-trivial calibration. **Manakul et al. [9]** (SelfCheckGPT) detect hallucination by sampling multiple responses and measuring consistency, requiring no internal access but paying $k\times$ generation cost. **Kuhn et al. [10]** introduced semantic entropy, clustering sampled generations by bidirectional NLI entailment before computing entropy over *meanings* rather than surface strings; **Farquhar et al. [11]** extended this and demonstrated it at scale. Semantic entropy is currently the strongest general-purpose baseline in this space and is the principal comparison target of this project.

4.3 Representational similarity and alignment

The cross-model experiment requires machinery from a different literature. **Kornblith et al. [12]** introduced Centered Kernel Alignment (CKA) for comparing representations across networks of differing width, and **Raghu et al. [13]** earlier proposed SVCCA for the same purpose. **Schönemann [14]** gives the closed-form solution to the orthogonal Procrustes problem, and **Conneau et al. [15]** demonstrated that Procrustes alignment of independently trained embedding spaces supports cross-lingual transfer — the closest existing precedent for the mapping this project applies between model families. To our knowledge these tools have not been combined with truthfulness probing.

4.4 Positioning

Work	Scale	Setting	Cross-model?	Cross-lingual?	Cost analysis?
Azaria & Mitchell [1]	6B–30B	Statements	No	No	No
Burns et al. [2]	$\geq 1.5B$	Statements	No	No	No
Li et al. [3]	7B–13B	QA	No	No	No
Marks & Tegmark [4]	7B–70B	Statements	No	No	No
Chen et al. [5]	7B–13B	Generation	No	No	Partial
Orgad et al. [7]	7B–8B	Generation	No	No	No
This project	1B–4B	Both	Yes	Yes	Yes

Table 4. Positioning against prior work. Empty cells in the upper rows are the project's opportunity.

5 Research Gaps

- G1 — The small-model regime is unstudied.** Published probing results concentrate on 7B–70B models. Whether a clean linear truthfulness direction exists at 1B–3B is genuinely unknown, and it is precisely the regime relevant to on-device and cost-constrained deployment. A negative result here is a scientifically meaningful statement about emergence.
- G2 — Probes are assumed to be model-specific, never tested.** Every probing paper trains and evaluates within a single model. The question of whether a truthfulness detector is *portable* across independently trained model families has, to our knowledge, not been posed — largely because the representation spaces are not directly comparable. That obstacle is surmountable with alignment techniques borrowed from the representational-similarity literature (Section 4.3).
- G3 — Cross-lingual encoding of truthfulness is unexamined.** Multilingual models must represent the same fact expressed in different languages. Whether the *truthfulness* feature is shared across languages, or re-learned per language, is open.
- G4 — Evaluation ignores the deployment question.** AUROC alone does not answer "how much does error drop if the system abstains on 20% of inputs?" The selective-prediction literature provides the correct instruments (risk-coverage curves, AURC), and they are almost never applied in probing papers. Nor is detection quality plotted against inference cost, despite cost being the entire practical argument for internal-state methods.

6 Research Questions and Hypotheses

ID	Research question	Hypothesis	Falsification criterion
RQ1	Does a linearly decodable truthfulness signal exist in sub-4B models, and where?	H1: Yes; AUROC peaks at 55–80% relative depth, on exact-answer token positions	Best linear probe fails to exceed log-probability baseline by a statistically significant margin
RQ2	Does a probe trained on model A transfer to model B under a learned linear alignment?	H2: Aligned transfer substantially exceeds a random-map control, while remaining below in-model performance	Aligned transfer is statistically indistinguishable from the random-orthogonal control
RQ3	Does an English-trained probe transfer to Indic-language prompts on the same model?	H3: Partial transfer, correlating with the model's multilingual competence	Cross-lingual AUROC collapses to chance across all tested models
RQ4	What is the detection-quality-per-FLOP frontier?	H4: Probes approach semantic entropy's AUROC at ~10× lower inference cost	Probes are dominated by a cheap baseline on both axes simultaneously

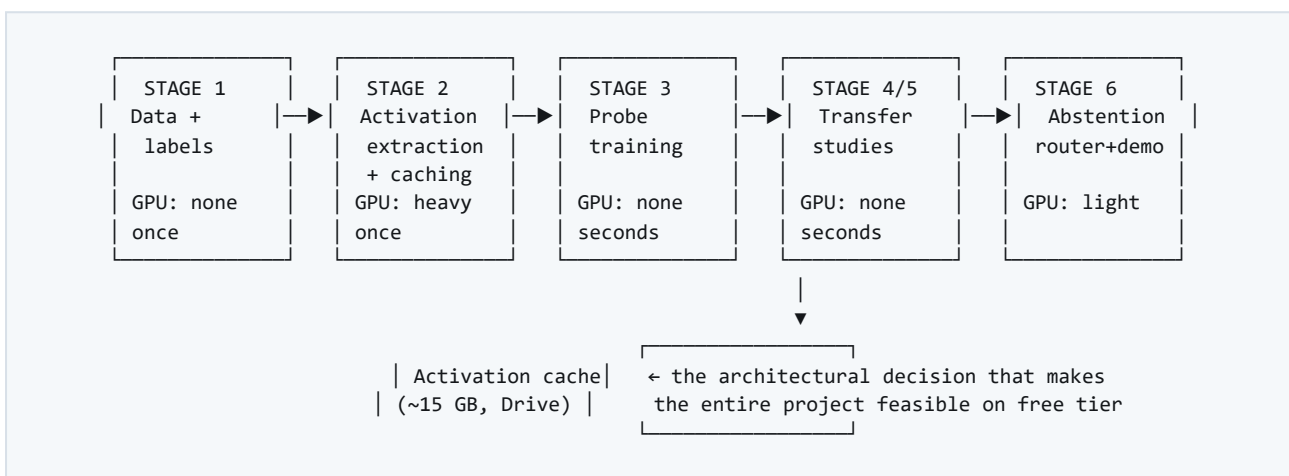
Table 5. Research questions with explicit falsification criteria stated in advance.

DESIGN PROPERTY: THE PROJECT CANNOT PRODUCE A NULL OUTCOME

Each hypothesis has been formulated so that its refutation is itself a publishable finding. If H1 fails, the result is *"linear truthfulness representation does not emerge below 4B parameters"* — a statement about scaling. If H2 fails, the result is *"truthfulness directions are model-idiosyncratic"*, supported by CKA analysis explaining why. This property is the principal reason this topic was selected for a time-boxed degree project.

7 Proposed Methodology

7.1 System architecture



The pipeline is deliberately decomposed so that the only GPU-expensive stage runs exactly once. Stages 3–5 — which contain all the scientific content and all the iteration — read from a cached tensor store and execute on CPU in seconds. Each stage is independently testable and has a well-defined interface: Stage 1 emits `(prompt, generation, label)` records; Stage 2 emits `(record_id, layer, position) → vector`; Stages 3–6 consume only that.

7.2 Stage 1 — Datasets and label construction

Two labelling regimes are used, and the distinction between them is the most important methodological point in this document.

Regime A — statement-level truthfulness

The model is shown a declarative statement of known truth value; the probe reads activations from the forward pass over that statement. Labels are supplied by the dataset. This regime is fast, noise-free, and used to validate the pipeline and to fit the alignment maps in Stage 4.

Regime B — self-generated hallucination

The model is asked a question, *generates its own answer*, and the answer is scored against gold references. The probe reads activations from the model's own generative forward pass. The label therefore describes *this model's own error on this specific input*.

WHY REGIME B MUST BE THE PRIMARY RESULT

Regime A demonstrates only that *a classifier can judge a sentence*. Regime B demonstrates that *a model's internal state predicts its own failure* — a strictly stronger and scientifically meaningful claim. Reviewers familiar with this literature check for exactly this distinction; papers that report Regime A results while making Regime B claims are routinely rejected. Regime A is scaffolding. Regime B is the paper.

Dataset	Regime	Size used	Label source	Role
SAPLMA [1]	A	~6k	Dataset annotation	Pipeline validation; layer sweep; alignment fitting
FEVER [16]	A	~10k	SUPPORTS / REFUTES	Statement-regime scale-up
TruthfulQA [17]	A / B	817	MC1 / MC2 options	Adversarial misconception subset
TriviaQA [18]	B	~12k	Normalised EM + token-F1 vs. gold aliases	Primary generative benchmark
HotpotQA [19]	B	~8k	Normalised EM + token-F1	Multi-hop; harder retrieval
HaluEval [20]	B	~10k	Provided hallucinated / correct pairs	Cross-task transfer (QA / dialogue / summarisation)
XSum-faithfulness [21]	B	~2k	Human faithfulness annotations	Faithfulness vs. factuality contrast
Indic QA subset	B	~4k	EM/F1 on translated gold answers	Cross-lingual transfer (RQ3)

Table 6. Dataset inventory. All sources are publicly downloadable without credentialing or licence application.

Correctness labelling for Regime B

Automatic labelling is performed by normalised exact match against the full alias set, plus token F1 with a threshold. This is standard but imperfect: paraphrases and equivalent surface forms generate false negatives. Two mitigations are built into the plan:

- 1. Manual audit.** A stratified sample of 200 labelled generations is hand-checked; inter-annotator agreement (Cohen's κ) and the estimated automatic-labelling error rate are reported in the paper.
- 2. Ambiguity band.** Examples whose F1 falls in a middle band ($\tau_{lo} < F1 < \tau_{hi}$) are excluded from the primary evaluation and reported separately, so that label noise does not silently inflate or deflate results.

7.3 Stage 2 — Activation extraction and caching

Models are loaded in 4-bit NF4 quantisation via `bitsandbytes` [22] and run in inference mode only. Activations are captured with forward hooks registered on each decoder block, or equivalently via `output_hidden_states=True`.

Extraction axes

Axis	Values	Rationale
Layer ℓ	All L layers (26–36 depending on model)	Produces the depth profile; tests H1
Token position t	(a) last prompt token; (b) last generated token; (c) exact-answer tokens ; (d) mean over answer span	[7] reports (c) is strongest — independent verification at small scale
Granularity	Residual stream; optionally per-attention-head outputs	Head-level probing (ITI-style [3]) yields a head-localisation map

Table 7. Extraction sweep. The full grid is captured in a single pass over the data.

Storage analysis

For a model with $L = 30$ layers and $d = 2048$, storing all layers at one token position in `float16` costs $30 \times 2048 \times 2 \text{ B} \approx \mathbf{123 \text{ KB per example}}$. For 20,000 examples and 4 token positions this is approximately 9.8 GB per model. With five models the cache approaches 50 GB, which exceeds a standard Drive allocation. Three mitigations are applied:

- **Layer subsampling after the first sweep.** Once the depth profile is measured on one model, retain only every second layer plus a dense band around the peak — a $\sim 3\times$ reduction.
- **Position prioritisation.** Retain full-grid positions for two models; retain only the best-performing position for the remaining models.
- **Shard-and-stream.** Write `float16` shards of 512 examples, memory-mapped on read, so no experiment requires the full cache resident in RAM.

```
cache/  
  {model_id}/  
    manifest.json          # record ids, layer index, position codes, dtype, shapes  
    shard_00000.npy        # (512, n_layers, n_positions, d) float16  
    shard_00001.npy  
    ...  
    labels.parquet         # record_id, dataset, topic, question, generation, gold, em, f1, label
```

THE DECISION THAT MAKES THIS PROJECT FEASIBLE

Extraction is executed **once**. Every subsequent experiment — every layer, every probe family, every transfer configuration, every ablation — is a NumPy/scikit-learn operation over cached tensors, running in seconds on CPU. This converts a study that superficially requires sustained GPU access into one that requires a few days of GPU time total, and it is the reason the experimental matrix in Section 10 is affordable on a free tier.

7.4 Stage 3 — Probe families

Probe	Form	Purpose in the study
Logistic regression	$\sigma(\mathbf{w}^\top \mathbf{h} + b)$, L2-regularised	Primary. Linearity is the scientific claim; $\mathbf{w}$ is directly interpretable
Mass-mean difference	$\mathbf{w} = \mu_{\text{false}} - \mu_{\text{true}}$, optionally whitened by Σ^{-1}	Training-free; robust under label noise and small samples; strong sanity baseline
MLP probe	One hidden layer (256 units, ReLU)	Quantifies how much non-linearity buys — and flags probe-expressivity confounds
CCS (unsupervised)	Consistency + confidence loss over contrast pairs [2]	Tests whether the direction exists <i>without</i> supervision
Multi-layer ensemble	Concatenate top- k layers, then linear probe	Tests whether the signal is distributed across depth

Table 8. Probe families. Rows 1–2 carry the primary claims; rows 3–5 are controls and extensions.

7.5 Stage 4 — Cross-model transfer (RQ2, the central novelty)

The obstacle

Independently trained models occupy incomparable representation spaces. Llama-3.2-1B has $d = 2048$, Qwen2.5-1.5B has $d = 1536$, Gemma-2-2B has $d = 2304$. Even at equal width there is no reason for basis vectors to correspond. A probe vector cannot simply be carried across.

The method

Present the *same* N stimuli to both models and collect paired activation matrices $\mathbf{A} \in \mathbb{R}^{N \times d_A}$ and $\mathbf{B} \in \mathbb{R}^{N \times d_B}$. Because rows are paired by stimulus, a mapping can be fitted. Three variants are compared:

(i) *Ridge regression — unconstrained linear map.*

$$\mathbf{W}^* = \operatorname{argmin}_{\mathbf{W}} \|\mathbf{A}\mathbf{W} - \mathbf{B}\|_F^2 + \lambda \|\mathbf{W}\|_F^2 \Rightarrow \mathbf{W}^* = (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{B} \quad (4)$$

(ii) *Orthogonal Procrustes — rotation only* (after PCA-matching dimensions to $r = \min(d_A, d_B)$). With $\mathbf{A}^\top \mathbf{B} = \mathbf{U}\Sigma\mathbf{V}^\top$ the closed-form solution [14] is:

$$\mathbf{R}^* = \operatorname{argmin}_{\mathbf{R}^\top \mathbf{R} = \mathbf{I}} \|\mathbf{A}\mathbf{R} - \mathbf{B}\|_F^2 = \mathbf{U}\mathbf{V}^\top \quad (5)$$

(iii) *CCA / SVCCA subspace alignment* [13] — align maximally correlated subspaces rather than the full space.

Transfer is then evaluated by applying probe g_A , trained on model A, to mapped activations of model B — where the map is fitted on stimuli disjoint from the probe's training set and from the evaluation set.

Controls that make the result defensible

- 1. Random-orthogonal control.** Replace $\mathbf{R}^*$ with a random orthogonal matrix. Transfer performance must collapse; if it does not, the "transfer" is an artefact of class priors rather than representational correspondence. *This is the single most important control in the project.*

2. **Unaligned control.** Apply g_A to raw **B** activations (zero-padded or truncated). Expected to be at chance.
3. **Upper bound.** A probe trained natively on model B, establishing the ceiling that aligned transfer is measured against.
4. **Similarity predictor.** Compute layer-wise CKA [12] between the two models and test whether CKA predicts transfer success — converting a bare empirical number into an explanation.

The output is a $K \times K$ transfer matrix over model pairs and layer pairs, reported alongside its CKA counterpart.

7.6 Stage 5 — Cross-lingual transfer (RQ3)

A subset of factual questions is rendered in Hindi and one additional Indic language (Telugu or Bengali), using a public machine-translation system with a manually verified sample to bound translation quality. A probe is trained on English activations and evaluated on activations from the *same model* processing the translated prompts. Three conditions are compared: train English / test English; train English / test Indic; train Indic / test Indic. Results are analysed against each model's independently measured multilingual competence, so that a transfer failure can be attributed either to language-specific truthfulness encoding or to the model simply not knowing the fact in that language — a confound that must be separated and will be.

7.7 Stage 6 — Selective prediction and abstention

The probe score is thresholded to produce an abstention policy: answer if $s(x) < \tau$, otherwise return "I am not confident". Sweeping τ traces the risk–coverage curve, the standard instrument of the selective-prediction literature [23, 24]:

$$\text{coverage}(\tau) = 1/n \sum_i \mathbf{1}[s_i < \tau], \quad \text{risk}(\tau) = \sum_i c_i \mathbf{1}[s_i < \tau] / \sum_i \mathbf{1}[s_i < \tau] \quad (6)$$

The deliverable is a demonstration application in which the model answers a question while a confidence indicator, driven purely by its internal state, is displayed alongside — flagging likely hallucinations before any external verification occurs. Measured wall-clock overhead of the probe is reported against measured overhead of each sampling-based baseline.

8 Baselines

The credibility of the entire project rests on this section. A probe that is not compared against strong, correctly implemented baselines is not evidence of anything.

Baseline	Definition	Extra fwd passes	Notes
Mean / min token log-prob	Average or minimum log p over answer tokens	0	Trivial but surprisingly competitive; must be reported
Predictive entropy	Token-level entropy averaged over the answer span	0	Conflates linguistic and factual uncertainty
P(True) [8]	Re-prompt the model to judge its own answer	1	Cheap; calibration typically weak at small scale
SelfCheckGPT [9]	k samples; measure inter-sample consistency	$k=5-10$	Strong; expensive
Semantic entropy [10, 11]	k samples clustered by bidirectional NLI entailment; entropy over meaning clusters	$k=10 + \text{NLI}$	Principal comparison target. DeBERTa-MNLI used as the entailment model
EigenScore (INSIDE) [5]	log-det of internal-state covariance across samples	$k=10$	Internal-state method that still pays sampling cost

Table 9. Baseline suite. Every baseline is evaluated on identical splits with identical labels.

THE WIN CONDITION

Semantic entropy is expected to remain the strongest detector in absolute AUROC. The claim being tested is *not* that the probe beats it outright, but that the probe approaches it at roughly one order of magnitude less inference cost. The headline figure is therefore two-dimensional — AUROC against measured inference FLOPs — with the probe occupying the upper-left region of the plot. Framing the contribution as a *cost-quality frontier* rather than a leaderboard win is both more honest and more defensible under review.

9 Evaluation Protocol and Metrics

9.1 Metrics

Metric	Definition	Why it is included
AUROC	Area under the ROC curve of the hallucination score	Threshold-free; comparable to prior work
AUPRC	Area under precision–recall curve	Hallucinations are the minority class; AUROC alone is optimistic
AURC	Area under the risk–coverage curve	The deployment-relevant summary statistic
Accuracy @ 80% coverage	Accuracy on retained examples after abstaining on 20%	Directly interpretable; the number to quote in a talk
ECE	Expected calibration error over 10 bins	Are the probe's probabilities meaningful, not merely ranked correctly?
Inference overhead	Measured wall-clock ms and estimated FLOPs per detection	The entire practical argument for the method

Table 10. Evaluation metrics.

9.2 Data splitting discipline

NON-NEGOTIABLE

Splits are constructed **by topic and by entity**, never at random. Random splitting places the same entity in both train and test, and the probe learns entity frequency rather than truthfulness — inflating AUROC by a wide and misleading margin. This is the most common methodological failure in the probing literature and the most common reviewer objection. The splitting procedure will be described explicitly in the paper, and a random-split ablation will be reported alongside the grouped-split result to quantify exactly how much leakage inflates the number.

9.3 Statistical reporting

- Every headline number reported as mean $\pm$ std over 5 seeds.
- 95% confidence intervals via bootstrap resampling (1,000 replicates) on the test set.
- Paired comparisons against baselines via bootstrap over the same resamples; differences reported with CIs rather than bare p -values.
- All hyperparameters (regularisation strength, layer choice, threshold) selected on validation data only. Test data is touched once, at the end.

9.4 Negative controls

1. **Label shuffling.** Retrain probes on permuted labels; AUROC must fall to 0.5. Detects leakage in the pipeline.

2. **Random-projection control.** Probe on randomly projected activations of matched dimension — bounds how much performance comes from dimensionality alone.
3. **Untrained-model control.** Extract from a randomly initialised model of identical architecture. Any above-chance result indicates a confound in the input encoding rather than learned knowledge.
4. **Random-orthogonal alignment control** (Section 7.5).

10 Experimental Matrix and Planned Results

10.1 Models

Model	Params	d	L	Family	Role
Llama-3.2-1B-Instruct	1.24B	2048	16	Llama	Lower scale bound
Llama-3.2-3B-Instruct	3.21B	3072	28	Llama	Within-family scale contrast
Qwen2.5-1.5B-Instruct	1.54B	1536	28	Qwen	Cross-family; multilingual
Gemma-2-2b-it	2.61B	2304	26	Gemma	Cross-family; third architecture
Phi-3-mini-4k-instruct	3.82B	3072	32	Phi	Upper scale bound (4-bit)

Table 11. Model inventory. Architectural details to be re-verified against config files at implementation time. Five models across four families gives a 4×4 cross-family transfer matrix plus a within-family scale control.

10.2 Planned figures and tables

Ref	Content	Question it answers
Fig. 1	AUROC heatmap: layer × token position, per model	Where does the signal live? (RQ1)
Fig. 2	AUROC vs. measured inference cost, all methods	The cost–quality frontier (RQ4) — <i>headline figure</i>
Fig. 3	Risk–coverage curves, probe vs. baselines	Deployment value of abstention
Fig. 4	Cross-model transfer matrix, with random-map control panel	Are probes portable? (RQ2) — <i>novelty figure</i>
Fig. 5	Layer-wise CKA overlaid on transfer success	Why does transfer succeed or fail?
Fig. 6	Cross-lingual transfer bars vs. multilingual competence	Is truthfulness language-independent? (RQ3)
Fig. 7	Attention-head localisation map	Is the signal sparse across heads?
Tab. A	Main results: all models × datasets × methods	Primary quantitative claim
Tab. B	Ablations: probe family, layer choice, split strategy, sample size	Robustness of the claim
Tab. C	Error-type breakdown (per Table 3 taxonomy)	What the probe can and cannot detect

Table 12. Planned result artefacts, defined in advance of experimentation.

11 Implementation Plan

11.1 Software stack

Component	Tooling	Purpose
Model execution	PyTorch, HuggingFace transformers , accelerate	Loading, generation, hidden-state capture
Quantisation	bitsandbytes (NF4, 4-bit) [22]	Fit 3–4B models in 15 GB VRAM
Datasets	datasets , pandas , pyarrow	Loading, splitting, label storage
Probes & stats	scikit-learn , numpy , scipy	Logistic regression, ridge, SVD, Procrustes, CCA, bootstrap
Entailment	DeBERTa-v3-large-MNLI	Semantic-entropy clustering
Translation	IndicTrans2 or equivalent public system	Cross-lingual dataset construction
Tracking	Weights & Biases (free tier) or CSV logs	Experiment provenance
Demo	Gradio	Interactive abstention demonstration
Paper	LaTeX (IEEE conference template), Overleaf	Manuscript preparation

Table 13. Software stack. Every component is free and requires no institutional licence.

11.2 Repository structure

```
hallucination-probes/
├─ configs/           # YAML: model specs, dataset specs, experiment definitions
├─ src/
│   ├─ data/          # loaders, prompt templates, EM/F1 scoring, topic-grouped splitters
│   ├─ extract/        # hook registration, batched generation, shard writer
│   ├─ probes/         # logistic, mass-mean, MLP, CCS, ensemble
│   ├─ transfer/       # ridge, Procrustes, CCA, CKA, random-map controls
│   ├─ baselines/      # logprob, entropy, P(True), SelfCheckGPT, semantic entropy, EigenScore
│   ├─ eval/           # AUROC, AUPRC, AURC, ECE, bootstrap CIs, risk-coverage
│   └─ viz/            # figure generation, one module per planned figure
├─ notebooks/        # Colab entry points, one per pipeline stage
├─ tests/             # unit tests: shape contracts, label scoring, split disjointness
├─ demo/              # Gradio abstention app
└─ paper/             # LaTeX source, generated figures
```

Two engineering disciplines are enforced from week 1. First, **resumability**: extraction writes shards incrementally with a manifest, so a Colab disconnection costs at most one shard rather than one run. Second, **split-disjointness tests**: an automated test asserts that no topic or entity appears in more than one split, so the leakage failure described in Section 9.2 is structurally prevented rather than merely intended against.

11.3 Compute and storage budget

Activity	Est. GPU-hours	Notes
Regime A extraction (5 models)	~8	Forward passes only, batched
Regime B generation + extraction	~35	Dominant cost; generation is sequential
Baseline sampling ($k=10$, subset)	~25	Restricted to a fixed evaluation subset to control cost
Cross-lingual extraction	~10	Two languages, subset of models
Probe training and all transfer studies	0	CPU only , seconds per experiment
Total	~78 GPU-hours	Spread over 24 weeks ≈ 3.3 h/week — comfortably within free-tier limits

Table 14. Compute budget. Storage: ~15 GB after the layer-subsampling reduction of Section 7.3.

11.4 Reproducibility

- All seeds fixed and recorded; all configs versioned in git.
- Dataset checksums recorded in the manifest to detect silent upstream changes.
- Code released publicly under a permissive licence on acceptance.
- The activation cache manifest published so that probe experiments can be replicated without re-running extraction.

12 Timeline and Milestones

Weeks	Phase	Activities	Exit criterion (verifiable)
1–2	Foundation	Colab environment; 4-bit loading; forward hooks; shard writer; reproduce SAPLMA on one model, one dataset	AUROC > 0.80 on Regime-A statements, grouped split
3–4	Sweep	Extend to all 5 models; full layer × position sweep; build cache	Fig. 1 produced; peak layer/position identified per model
5–7	Regime B	Generation pipeline; EM/F1 labelling; manual audit of 200 samples; ambiguity-band filtering	Labelled generative dataset; κ reported; probes trained on self-generated labels
8–10	Baselines	Implement all six baselines; measure wall-clock and FLOPs for each	Fig. 2 and Fig. 3 produced; Table A first draft
11–14	Transfer	Paired-stimulus collection; ridge / Procrustes / CCA; all four controls; CKA analysis	Fig. 4 and Fig. 5 ; RQ2 answered either way
15–17	Cross-lingual	Translation + verification; Indic extraction; three-condition comparison	Fig. 6 ; RQ3 answered with confound controlled
18–19	Application	Abstention router; Gradio demo; latency benchmarking	Working demo; overhead table complete
20–22	Consolidation	Ablations; negative controls; error-type breakdown; failure analysis	Tables B and C complete; all controls pass
23–24	Dissemination	Manuscript; internal review; arXiv preprint; conference submission	Paper submitted ; code released

Table 15. 24-week plan. Weeks 11–14 carry the novelty; weeks 1–10 exist to make those four weeks trustworthy.

SCHEDULING NOTE

The two highest-risk dependencies are front-loaded deliberately. If the Regime-A pipeline does not reach AUROC > 0.80 by week 2, the small-model hypothesis (H1) is already in doubt and the project pivots — while 22 weeks remain — toward the scale-emergence framing described in Section 14. No such decision is deferred past week 7.

13 Deliverables

#	Deliverable	Description
D1	Research paper	6–8 page manuscript, IEEE conference format, submitted to a Scopus-indexed venue
D2	arXiv preprint	Posted at submission time — citable immediately, independent of review outcome
D3	Open-source codebase	Documented, tested, reproducible; per Section 11.2
D4	Activation-cache manifest	Enables replication of all probe experiments without GPU access
D5	Interactive demo	Gradio application showing live internal-state confidence and abstention
D6	Project report	Full academic report meeting university requirements
D7	Presentation	Viva/defence deck, plus a 5-minute version for interviews

Table 16. Deliverables.

14 Risk Analysis and Contingencies

Risk	Likelihood	Impact	Mitigation / contingency
Signal too weak at 1B scale	Medium	Medium	3B and 3.8B models included. If small models genuinely fail, the paper is reframed as a scale-emergence study — "linear truthfulness representation does not emerge below $\mathcal{X}\mathcal{B}$ " — which is publishable in its own right
Cross-model transfer fails (H2 refuted)	Medium	Low	Pre-planned as a publishable negative result, supported by CKA evidence. The narrative becomes "truthfulness directions are model-idiosyncratic, and here is the representational reason"
EM/F1 label noise	High	Medium	Manual audit with κ reporting; ambiguity-band exclusion; sensitivity analysis across thresholds
Colab disconnection / quota exhaustion	High	Low	Resumable sharded extraction from week 1; Kaggle (30 h/week GPU) as a parallel account; extraction is the only GPU-bound stage
Storage exceeds Drive quota	Medium	Medium	Layer subsampling and position prioritisation (Section 7.3); <code>float16</code> throughout; per-model caches deletable after probing
Reviewer: "this replicates SAPLMA"	Medium	High	Transfer axes, selective-prediction protocol, and cost–quality framing must appear <i>in the abstract</i> . Abstract is drafted in week 10, before the transfer experiments, and used to keep the narrative disciplined
Semantic-entropy baseline too costly to run fully	Medium	Low	Evaluate on a fixed, pre-registered subset; report subset size and justify; keep all methods on the identical subset for fairness
Translation quality confounds RQ3	Medium	Medium	Manual verification of a translated sample; report translation quality; separate "model lacks the fact in this language" from "probe fails to transfer" via a per-language accuracy control

Table 17. Risk register. Note that the two scientific risks both have publishable failure modes by design.

15 Expected Contributions and Publication Plan

15.1 Claimed contributions

1. **C1.** The first systematic characterisation of linearly decodable truthfulness signals in sub-4B language models, across five models, four families, and both statement and generative regimes.
2. **C2.** The first study of *cross-model-family probe transfer* under learned representational alignment, with rigorous controls and a CKA-based account of when transfer succeeds.
3. **C3.** The first evaluation of cross-lingual transfer of truthfulness probes to Indic languages, with the multilingual-competence confound explicitly separated.
4. **C4.** A selective-prediction evaluation protocol that reports detection quality against measured inference cost, establishing a cost–quality frontier for hallucination detection at small scale.
5. **C5.** An open, reproducible codebase and activation-cache manifest that lowers the barrier for subsequent work on consumer hardware.

15.2 Venues

Tier	Venues	Notes
Primary target	IEEE INDICON, IEEE TENCON, IEEE CONECCT, IEEE ANTS, ICCCNT	Scopus-indexed; timelines compatible with a 24-week plan
Stretch	TrustNLP (ACL workshop); reliable-ML / interpretability workshops at NeurIPS, ICLR	Attempt only if RQ2 yields a strong positive or a clean, well-explained negative
Immediate	arXiv (cs.CL)	Post at submission — citable and timestamped regardless of review outcome

Table 18. Publication targets.

15.3 Note on venue selection

Conference deadlines are the binding constraint on the timeline, and they move year to year. Actual dates for the target venues must be verified against official calls for papers at the start of week 1, and the schedule in Table 15 adjusted to the nearest suitable deadline. Predatory venues — those soliciting by email, promising review in days, or charging fees disproportionate to their indexing — should be avoided regardless of claimed indexing status; an arXiv preprint is strictly more valuable than a publication in a disreputable venue.

16 Ethical Considerations and Limitations

16.1 Dual use

A detector that identifies the model's internal truthfulness signal also identifies a direction that could, in principle, be manipulated to make hallucinations *harder* to detect. This is a known property of

interpretability work. The risk is judged low relative to the defensive benefit, and no adversarial evasion method will be developed or released as part of this project.

16.2 Overclaiming and automation bias

A confidence indicator that is wrong in a confident direction is worse than no indicator, because users over-trust visible confidence signals. The demonstration application will present the probe output as an *advisory* signal with its measured false-negative rate displayed alongside, and the paper will state operating-point-specific error rates rather than a single headline number.

16.3 Stated limitations

- Results are established for models below 4B parameters and may not extrapolate upward.
- Short-form factual QA is the primary setting; long-form multi-claim generation is not addressed.
- Automatic correctness labels carry residual noise, bounded but not eliminated by manual audit.
- Cross-lingual results depend on machine-translated prompts and inherit translation artefacts.
- The probe detects errors *this model* makes on *this distribution*; robustness to adversarial or heavily out-of-distribution prompts is untested.

17 Glossary

Term	Plain-language meaning
Hallucination	The model states something false with full confidence
Activation / hidden state	The internal numbers the model computes while processing text, before producing words
Residual stream	The shared "notepad" each layer reads from and writes to; the source of the numbers we probe
Probe	The small classifier trained on those numbers — in the primary case, logistic regression
Linear probe	A probe that multiplies the activation by one learned vector and adds a number. Deliberately simple
AUROC	Detector quality score. 0.5 = coin flip, 1.0 = perfect. Above ~0.75 is useful
Baseline	Existing simpler methods that must be beaten, or matched at lower cost, for the work to count
Abstention	Letting the system answer "I am not confident" instead of guessing
Risk–coverage curve	A plot of error rate against how many questions the system chooses to answer
Cross-model transfer	Training the detector on one AI model and testing whether it still works on a different one
Procrustes alignment	Finding the rotation that best lines up two different models' internal coordinate systems
CKA	A similarity score between two models' internal representations
Semantic entropy	Strongest existing baseline: sample the model ten times and measure meaning-level disagreement
Quantisation (4-bit)	Compressing model weights so a 3B model fits in a free Colab GPU
Grouped split	Splitting data by topic rather than randomly, so the probe cannot cheat by memorising entities

Table 19. Glossary of terms used in this document.

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Student — signature and date

Project guide — signature and date